

Suggested Teaching Guideline for
Cyber Forensics
PGCP-ITISS February 2026

Duration: 20 Theory hrs + 20 Lab hrs + 20 Self-Learning hrs (Total: 60 hrs)

Objective: To introduce the student to Cyber Forensics and cyber laws.

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment – 20% weightage

List of Books / Other training material

Courseware: No specific courseware for the module, faculty may share some course materials.

Reference:

- Guide to Computer Forensics and Investigations Computer Forensics and Investigations/nelson
- Digital Forensic: For Mumbai University by Dr. Nilakshi, Dr. Dhananjay

Note: Conduct lab sessions based on case studied. Faculty can ask students to get some casestudies and choose among that for explanation.

Session 1: (2T + 4SL)

Theory:

Introduction

- Overview - Computer Forensics
- Difference – Computer Crime & Un-authorized activities
- Cyber Laws
- Process of Computer Forensics (six)
- Need for forensics investigator

Self-Learning: Note: This needs to be completed before beginning of the Session 2.

- Learn about File system
- Find difference between logical vs physical storage
- Learn about metadata
- Study about Timestamps
 - ◆ Created
 - ◆ Modified
 - ◆ Accessed
- Understand about Timestamps manipulation
- Find out difference between FAT32 & NTFS based on the following:

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- ◆ File allocation table vs MFT (Master File Table)
- ◆ How files are linked vs MFT entry
- ◆ How deleted files are recovered (concepts)
- Find out what happens when a file is deleted
- Find out about File deletion ≠ data erased
- Learn about: Slack Space, Unallocated space
- Find out about file signature
- Study about common signatures like:
 - ◆ JPG, PNG, PDF, DOCX, EXE
- Create -> Modify -> delete a file and observe timestamps
- Use Hex editor to view file header
- Go through the article for your reference:
- <https://youaccel.com/lesson/file-systems-and-their-significance-in-digital-forensics/premium?srsId=AfmBOooEYXK3hR6oP2QZE1qyC3B3HPRoLVEB1Ryulw7J1QSNgigerVdw>
- https://drive.google.com/file/d/12W_sBf2SwpgA6doppswyyxU-rLzYogMY/view?usp=drive_link
- <https://www.geeksforgeeks.org/operating-systems/understanding-file-system/>

Session 2: (2T + 4L + 8SL)

Theory:

- Computer Forensics Involves
 - ◆ Preservation
 - ◆ Identification
 - ◆ Extraction
 - ◆ Documentation
 - ◆ Interpretation
- Goals of Forensics Analysis
- Types of Cyber Forensics Techniques
- Cyber forensics Procedures
- Preparation
 - ◆ What to do before the incident
 - ◆ Incident response plan
 - ◆ Incident response team
 - ◆ Detecting Incidents
 - ◆ Chain of custody

Lab 1 & 2:

- Simulating Incident Response and Chain of Custody Handling

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Self-Learning:

Note: This needs to be completed before the beginning of Session 3- 4.

- Find out the difference between OS vs Application
- Difference between user mode vs kernel mode
- Learn about process and process states
- Find out about the parent-child relationship, pid, ppid
- Understand why malware uses threads
- Understand the difference between packet capture vs packet analysis
- Understand the difference between live capture vs PCAP analysis
- Learn about what a packet contains (Frames, headers, payload)
- Read about capture filters vs display filters
- Read about promiscuous mode
- Capture a traffic on your interface and apply the filters:
 - ◆ Ip.addr, tcp.port, udp.port, http, dns, icmp, tcp.flags
- Understand why system generates logs
- Find out the difference between event and log entry
- How logs are written to storage
- Understand local time vs UTC
- Learn about Epoch time, Time zones
- Learn about some event id's like:
 - ◆ Logon/logoff
 - ◆ Process creation
 - ◆ Account changes
 - ◆ Privilege escalation
- Go through the article:
- https://blog.cyber5w.com/network_forensics_with_wireshark
- <https://dr.lib.iastate.edu/server/api/core/bitstreams/8dd82e7d-2e78-4e2a-9bc9-cc9bc68c1fe0/content>

Session 3: (2T + 4L)

Theory:

- Evidence Checkout Log
- Handling Evidence
- First Response
- Formulate/Execute Response Strategy
- Forensic duplication
- Authenticate the Evidence
- Investigation
- Common Mistakes
- Detection

Lab 3 & 4:

- Demo and lab sessions on Evidence Checkout Log, Investigation and Detection.

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Session 4: (4T + 4SL)

Theory:

- The Initial Assessment
- Incident Notification Checklist
- Hexadecimal notation
- Practical Bits
- Slight diversion
- What is use of Hexadecimal

Self-Learning:

Note: This needs to be completed before beginning of Session 5-6.

- Learn about hashing before imaging vs hash after imaging
- Study about destination hash
- Study about hash comparison
- Learn about what happen when
 - ◆ Hash mismatched ignored
- Read about windows registry
- Learn about the following hives
 - ◆ SAM, SYSTEM, SOFTWARE, SECURITY, NTUSER.DAT
- Learn about ps, top, /proc
- Learn about /var, /log, auth.log, syslog, bash_history
- Learn about BIOS/UEFI
- Learn about boot sequence
- Find out about bootloader, OS loading
- Create a user & check logs
- Start/ Stop services and observe artifacts

Session 5: (2T + 2L)

Theory:

- Encoding And Encryption
- The Hex Editor
- Files
- Hashing
- Hashing DLs
- MD5 Hash collisions
- Hash Collisions
- Bit Rot

Lab 5:

- Demo and lab sessions on Encoding and Encryption.

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Session 6: (2T + 2L)**Theory:**

- Standard Operating Procedures
- Processing Crime and Incident Scenes – Working with Windows & DOS Systems Current

Lab 6:

- Processing a Crime Scene on Windows and DOS Systems

Session 7: (2T + 2L + 2SL)**Theory:**

- Forensics Implications
- Accreditation Standards
- Performing a Cyber Forensics Investigation
- Privacy and Cyber Forensics

Lab 7:

- Conducting a Cyber Forensics Investigation and Addressing Privacy Considerations

Self-Learning:**Note: This needs to be completed before the beginning of Session 8**

- Learn why imaging is required instead of working on originals
- Find out the difference between copying and forensic imaging
- Find out about different types of imaging:
 - ◆ Logical Imaging
 - ◆ Physical (Bit-by-Bit) Imaging
 - ◆ Live vs Dead Imaging
- Read about Common image formats and find out which tool supports what format
 - ◆ RAW (DD)
 - ◆ E01 (EnCase)
 - ◆ AFF
- Go through the article:
- https://drive.google.com/file/d/1aUeKxzi17n1nitzVNt9fSbJygtZ-tWu/view?usp=drive_link

Session 8: (2T + 4L)**Theory:**

- Computer Forensics Tools:
 - ◆ Sysinternals Suite
 - ◆ FTK Forensics Tool kit
 - ◆ FTK Imager
 - ◆ OSF
 - ◆ Hex

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Lab 8 & 9:

- Demo and lab sessions on Cyber Check Suit

Session 9 &10:(2T + 2L + 2SL)

Theory:

- Concept of collecting and analyzing artefacts from systems that are in the active state
- Concept of collection and analysis of artefacts from Linux systems
- An introduction to Mobile forensics

Lab 10:

- Demo and lab sessions on Live system forensics & Linux Forensics

Self-Learning: (SL is based on Session 8-10)

- Understand when file-level imaging is acceptable
- Understand imaging a single file
- Learn about what write blocking is
- Explore about Hardware write blockers, Software write blockers
- Learn about imaging errors & risks
- Learn about what happens when imaging is done without write blocker
- Read about what changes without write blocking
- Read about how court view evidence contamination